World's Top 2% Scientist — Stanford University, 2026

Dr. Shahnawaz Ahmad

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Employment Details

Aug, 2023 - Present — **Assistant Professor** in School of Computer Science Engineering and Technology at Bennett University, Greater Noida, India.
- **Subjects:** Machine Learning, Web Technologies, Data Structure, Cloud Computing, Design and Analysis of Algorithms
- **Responsibilities:** Faculties SCI (E) Publications Monitor
- **Member:** Anti Ragging Cell

Sep, 2015 - July, 2023 — **Assistant Professor** in the Department of Computer Science and Engineering at Gurugram University, Gurugram, Haryana, India.
- **Subjects:** Software Engineering, Web Technologies, Data Structure, Computer Network, Design and Analysis of Algorithms
- **Member:** Examination Cell, Scholarship Cell
- **Responsibilities:** Deputy Chair Data Center, Deputy Chair Student Outreach Programme

Aug, 2012 - July, 2013 — **Lecturer** in the Department of Computer Science and Engineering at Gautam Buddha Technical University, J. P. Nagar, Amroha, India.
- **Subjects:** Cloud Computing, Data Structure, Computer Network
- **Responsibilities:** Time Table Coordinator

Educational Qualifications

2023 — **Ph.D.** from *Faculty of Engineering and Technology, Jamia Millia Islamia University* (ranked 3rd among Indian universities (NIRF 2023, 2024) and awarded A++ by NAAC-India), New Delhi, India.
Percentage: **84.75%**
Thesis title: *Securing Cloud-Based Encryption Key Management Platform*
Supervisor: Prof. Shabana Mehfuz (Professor), JMI

2015 — **Master of Technology (M. Tech) in Computer Engineering** from *Maharshi Dayanand University*, Rohtak, Haryana, India.
Division: **First with Honours (79.90%)**
Dissertation title: *Recommender System for Software Requirements Negotiation and Prioritization*
Supervisor: Dr. Mohd Sadiq (Associate Professor), JMI

2012 — **Bachelor of Technology (B.Tech) in Computer Science and Engineering** from *Gautam Buddha Technical University*, Uttar Pradesh, India.
Division: **First with Honours (77.54%)**

2015 — **Certificate Course in Portuguese Language (Part-Time, 2015)** from *Jamia Millia Islamia University*, New Delhi, India

Research Publications

Journal Articles

1. Ahmad, S., Ansari, M., Mewada, A., Ahmed, G. et al. (2026). Blockchain-enhanced federated learning for IoT security and privacy using the GSR-C2N model. *Scientific Reports*.
2. Ahmad, S., Ansari, M. A., Mewada, A., Arif, M., & Yadav, A. S. (2026). Engineering interconnected cyber-physical-human systems for IoT applications: A review. *International Journal of System Assurance Engineering and Management*, 1–14.
3. Ahmad, S., Ansari, M. A., Mewada, A., & Hussain, S. (2026). Biocrypt-DNA: A DNA-inspired biocryptographic framework for secure cloud computing. *Life Cycle Reliability and Safety Engineering*, 1–20.
4. Ahmad, S., Hussain, S., Arif, M., & Ansari, M. A. (2026). Early-stage diabetes prediction using a stacked ensemble model enhanced with SHAP explainability. *Biomedical and Pharmacology Journal*, 19(1).
5. Ansari, S. A., Ahmad, S., Luqman, M., Ali, S., & Farooq, M. (2026). Gradient-free concatenated fusion neural network for early diagnosis of Parkinson's disease using MRI data. *Telecommunication Systems*, 89(3), 101.
6. Ansari, S. A., Luqman, M., Wajid, M. A., Wajid, M. S., & Ahmad, S. (2026). CoEDMTVideo: Collaborative edge computing and optimal ensemble deep Q-learning network for distributed model training in IoT video stream environments. *International Journal of Fuzzy Systems*, 1–25.
7. Chaturvedi, P., Ahmad, S., & Mewada, A. (2026). A comprehensive survey on fog computing: Architectures, techniques, challenges, and future directions. *Archives of Computational Methods in Engineering*, 1–43.
8. Chaturvedi, P., Ahmad, S., Nowroozi, E., & Rajarajan, M. (2026). Physiodrift-Fed: Confronting the drift-rate heterogeneity problem in federated learning for consumer cardiac wearables. *IEEE Transactions on Consumer Electronics*.
9. Mewada, A., Ahmad, S., & Ansari, M. A. (2026). ViT-HAR: Vision transformer-based human activity recognition in cluttered environments. *Sakarya University Journal of Computer and Information Sciences*, Advanced Online Publication, 190–204.
10. Mewada, A., Ansari, M. A., & Ahmad, S. (2026). Feature-driven deep architecture for high-precision recognition of complex handwriting. *Computational Intelligence*, 42(3), e70241.
11. Naz, R., Howlader, J., & Ahmad, S. (2026). A secure expert system framework for private function evaluation using functional encryption and multi-party computation. *Computers and Electrical Engineering*, 131, 110930.
12. Naz, R., Howlader, J., & Ahmad, S. (2026). Bridging functional encryption and secure function evaluation: Toward efficient and coercion-resistant multi-party systems. *Concurrency and Computation: Practice and Experience*, 38(18), e70949.
13. Nazim, M., Ahmad, S., Ansari, M. A., Arif, M., Ahmad, J., & Tanweer, S. (2026). A holistic strategy for choosing software requirements through multi-criteria intuitionistic fuzzy TOPSIS. *International Journal of System Assurance Engineering and Management*, 1–14.
14. Mustafa, J., Ahmad, S., Wasid, M., Maurya, N. K., & Ansari, M. A. (2026). Topological Synthesis of Planar Kinematic and Epicyclic Geared Mechanisms by Application of Total Distance Technique. *Journal of the Institution of Engineers India Series C*.

15 Mewada, A., Sharma, S. K., Ansari, M. A., Yadav, R., & Ahmad, S. (2026). Blockchain-Based Solutions for Enhancing Transparency and Efficiency in Organ Transplantation. *IEEE Communications Standards Magazine*, 10(1), 95–101.

16 Mustafa, J., Ahmad, S., Wasid, M., Ansari, M. A., & Ansari, S. A. (2026). Structural Synthesis of Epicyclic Gear Trains by Deep Learning and Generative AI for Adaptive Automation. *Journal of Computers Mechanical and Management*, 5(1), 25247.

17 Nazim, M., Ansari, M. A., Ahmad, S., & Arif, M. (2026). AI-based learning models for the life cycle prediction and detection of diabetes disorders: A comprehensive perspective. *Medical Data Mining*, 9(2), 11.

18 Mustafa, J., Ahmad, S., Maurya, N. K., Wasid, M., & Kumar, V. (2026). Supervised Machine Learning for Isomorphic Identification of Epicyclic Gear Trains. *Journal of the Institution of Engineers India Series C*, 107(3), 1425–1432.

19 Ansari, M. A., Mewada, A., Singh, D. K., Rusia, M. K., & Ahmad, S. (2026). ShopSafe: Artificial Intelligence-Powered Theft Prevention Using Multiscale Analysis. *Pattern Recognition and Image Analysis*, 36(2), 272–284.

20 Jaiswal, S., Mustafa, J., & Ahmad, S. (2026). Exploring metamaterials for healthcare: an extensive insight into capabilities and applications. *Bioinspired Biomimetic and Nanobiomaterials*, 15(3), 156–170.

21 Mubashra, Ansari, M. A., Srivastava, S., & Ahmad, S. (2026). A cross-dataset integrative strategy for identifying candidate diagnostic biomarkers in colorectal cancer. *Gene Reports*, 45, 102648.

22 Ahmad, S., Ansari, M., Arif, M., Ahmad, J., Mehfuz, S., Ogala, J. O., Adewale, O. et al. (2025). Hybrid recommender system for personalized healthcare: Integration of content-based and collaborative filtering approaches. *International Journal of System Assurance Engineering and Management*, 1–11.

23 Ahmad, S., Ansari, M., Mewada, A., Singh, P., Diwakar, M., Akhtar, S., Shivahare, B. D. et al. (2025). MPFAN: A novel multiscale network for brain tumor MRI classification. *The Open Bioinformatics Journal*, 18(1).

24 Ahmad, S., Arif, M., Ahmad, J., & Mehfuz, S. (2025). A TOTP-based secure data storage system in the cloud environment using the JWT token approach. *International Journal of System Assurance Engineering and Management*, 1–14.

25 Ahmad, S., Arif, M., Ansari, M., Nazim, M. et al. (2025). AI-driven fuzzy requirement engineering for cybersecurity: Integrating linguistic decision models. *Iran Journal of Computer Science*, 1–28.

26 Ahmad, S., Arif, M., Mehfuz, S., Ahmad, J., & Nazim, M. (2025). Deep learning-based cloud security: Innovative attack detection and privacy focused key management. *IEEE Transactions on Computers*, (99), 1–12.

27 Ahmad, S., Nazim, M., Arif, M., Ahmad, J., Mehfuz, S., & Ansari, M. A. (2025). Protecting data in the cloud: A systematic literature review of key management. *Concurrency and Computation: Practice and Experience*, 37(21-22), e70223.

28 Hussain, S., Ahmad, S., & Wasid, M. (2025). Artificial intelligence-driven intelligent learning models for identification and prediction of cardioneurological disorders: A comprehensive study. *Computers in Biology and Medicine*, 184, 109342.

29 Jayaswal, R., Ansari, M., Dixit, M., Singh, D. K., Ahmad, S. et al. (2025). Advances in facial expression recognition technologies for emotion analysis. *Discover Computing*, 28(1), 1–56.

30 Jayaswal, R., Ansari, M. A., Mewada, A., Ahmad, S., & Pathak, A. (2025). Catch before they fall: A pose-guided attention framework for indoor safety. *Iran Journal of Computer Science*, 1–14.

31 Sakib, M., Ahmad, S., Anwar, K., & Saqib, M. (2025). Optimizing support vector regression using grey wolf optimizer for enhancing energy efficiency and building prototype architecture. *Cluster Computing*, 28(1), 60.

32 Wasid, M., Abdullah Amar, A. Y., Ahmad, S., & Mustafa, J. (2025). Enhancing reliability in cybercrime detection using intelligent learning models for ensuring trustworthy social networks. *International Journal of Reliability, Quality and Safety Engineering*.

33 Ansari, M. A., Ahmad, S., & Mewada, A. (2025). Mitigating risk in medical AI: balancing X-ray datasets for reliable detection. *Life Cycle Reliability and Safety Engineering*.

34 Mustafa, J., Ahmad, S., & Hussain, S. (2025). Machine learning-based data processing for predictive modeling in mechanical systems. *Life Cycle Reliability and Safety Engineering*, 14(3), 445–454.

35 Okonkwo, U. C., Okafor, C. E., Ahmad, S., Ohagwu, C. C., Aronu, M. E., Okokpujie, I. P., Idumah, C. I. et al. (2025). Research advancements in machine learning-assisted design of reinforced composite radiation shields. *Life Cycle Reliability and Safety Engineering*, 14(3), 455–489.

36 Ansari, M. A., Ahmad, S., & Mewada, A. (2025). Multiscale parallel feature aggregation network with attention fusion (MPFAN-AF): A novel approach to cataract disease classification. *Medical Data Mining*, 8(4), 23.

37 Ahmad, S., Ansari, M. A., Mewada, A., & Arif, M. (2025). Advancing reliability and safety in IoT: engineering cyber-physical human systems for the future. *Life Cycle Reliability and Safety Engineering*, 14(4), 545–559.

38 Jayaswal, R., Ansari, M. A., Mewada, A., Pareek, P., & Ahmad, S. (2025). An in-depth exploration of structural pose estimation strategies and datasets. *Discover Computing*, 28(1), 222.

39 Wasid, M., Amar, A. Y. A., Ahmad, S., & Mustafa, J. (2025). Enhancing reliability and security in online social networks: intelligent learning models for cybercrime detection. *Life Cycle Reliability and Safety Engineering*, 14(4), 617–633.

40 Ahmad, S., Arif, M., Ahmad, J., Nazim, M., & Mehfuz, S. (2024). Convergent encryption enabled secure data deduplication algorithm for cloud environment. *Concurrency and Computation: Practice and Experience*, 36(21), e8205.

41 Ahmad, S., & Mehfuz, S. (2024). Efficient time-oriented latency-based secure data encryption for cloud storage. *Cyber Security and Applications*, 2, 100027.

42 Ahmad, S., Mehfuz, S., Urooj, S., & Alsubaie, N. (2024). Machine learning-based intelligent security framework for secure cloud key management. *Cluster Computing*, 27(5), 5953–5979.

43 Ahmad, S., Ogala, J. O., Ikpotokin, F., Arif, M., Ahmad, J., & Mehfuz, S. (2024). Enhanced CNN-DCT steganography: Deep learning-based image steganography over cloud. *SN Computer Science*, 5(4), 408.

44 Anwar, K., Goe, R., Ahmad, S., & Tripathi, S. (2024). Computational intelligence approaches for heart disease prediction: A comparative evaluation. *Biomedical and Pharmacology Journal*, 17(4), 2361–2380.

45 Ahmad, S., Mehfuz, S., & Beg, J. (2023). An efficient and secure key management with the extended convolutional neural network for intrusion detection in cloud storage. *Concurrency and Computation: Practice and Experience*, 35(23), e7806.

46 Ahmad, S., Mehfuz, S., & Beg, J. (2023). Hybrid cryptographic approach to enhance the mode of key management system in cloud environment. *The Journal of Supercomputing*, 79(7), 7377–7413.

47 Ahmad, S., Shakeel, I., Mehfuz, S., & Ahmad, J. (2023). Deep learning models for cloud, edge, fog, and IoT computing paradigms: Survey, recent advances, and future directions. *Computer Science Review*, 49, 100568.

48 Ogala, J. O., Ahmad, S., Shakeel, I., Ahmad, J., & Mehfuz, S. (2023). Strengthening KMS security with advanced cryptography, machine learning, deep learning, and IoT technologies. *SN Computer Science*, 4(5), 530.

49 Urooj, S., Lata, S., Ahmad, S., Mehfuz, S., & Kalathil, S. (2023). Cryptographic data security for reliable wireless sensor network. *Alexandria Engineering Journal*, 72, 37–50.

50 Okafor, C. E., Iweriolor, S., Ani, O. I., Ahmad, S., Mehfuz, S., Ekwueme, G. O., Chukwumuanya, O. E. et al. (2023). Advances in machine learning-aided design of reinforced polymer composite and hybrid material systems. *Hybrid Advances*, 2, 100026.

51 Ahmad, S., Mehfuz, S., & Beg, J. (2022). Assessment on potential security threats and introducing novel data security model in cloud environment. *Materials Today: Proceedings*, 62, 4909–4915.

52 Ahmad, S., Mehfuz, S., & Beg, J. (2022). Cloud security framework and key management services collectively for implementing DLP and IRM. *Materials Today: Proceedings*, 62, 4828–4836.

53 Ahmad, S., Mehfuz, S., Mebarek-Oudina, F., & Beg, J. (2022). RSM analysis based cloud access security broker: A systematic literature review. *Cluster Computing*, 25(5), 3733–3763.

54 Ahmad, S., Mehfuz, S., & Beg, J. (2020). A genetic algorithm based key management approach for enhancing data security in cloud environment. *International Journal of Advanced Trends in Computer Science and Engineering*, 9(5), 8943–8951.

55 Ahmad, S., & Sadiq, M. (2015). Recommender systems for software requirements negotiation and prioritization. *International Journal of Computer Applications*, 117(13).

Granted/Published Patents

1 Ahmad, S. (Sept 2021). *A new method of data sanitization for the privacy protection of cloud data in cloud computing.* Australian Govt. Patent No. 2021102595.

2 Ahmad, S. (Sept 2021). *A novel multilevel optimization for task scheduling and load balancing in cloud.* Australian Govt. Patent No. 2021103249.

3 Ahmad, S. (May 2021). *A system for resource aggregation to the cloud with optimal path detection.* Indian Patent Application No. 202141020723, CGPDTM, New Delhi.

4 Ahmad, S. (June 2021). *An efficient iris recognition system with complete local ternary pattern feature extraction and classification method.* Australian Govt. Patent No. 2021102309.

5 Ahmad, S. (Sep 2022). *Cyber attack prevention system for automotive system based on artificial intelligence.* South African Govt. Patent No. 2022/08301.

6 Ahmad, S. (May 2021). *Integrated solar panel as per sun direction using AI-based programming system.* Indian Patent Application No. 2021111017841, CGPDTM, New Delhi.

7 Ahmad, S. (May 2021). *System for control & manage artificial intelligent speaker functionality and operation using electronic pen.* Indian Patent Application No. 2021311016697, CGPDTM, New Delhi.

8 Ahmad, S. (Aug 2023). *Portable quick response (QR) code reader.* United Kingdom Design Patent No. 6303882.

9 Ahmad, S. (May 2021). *Secure long range device to communication methods for IoT devices using low power wide area network (LPWAN).* Australian Govt. Patent No. 2021102086.

10 Ahmad, S. (Sep 2023). *Self-adjusting webcam.* Indian Design Patent No. 380130-001, CGPDTM, New Delhi.

Authored Books

1. Shahnawaz Ahmad, S. H., Jiyaul Mustafa. (2024). *Machine Learning Applications in Mechanical Engineering.* Lap Lambert Academic Publishing, ISBN: 978-3-659-97728-2.
2. Shahnawaz Ahmad, J. B., Shabana Mehfuz. (2022). *Cloud Computing.* GCS Publishers, ISBN: 978-93-94304-10-9.
3. Shahnawaz Ahmad, J. B., Shabana Mehfuz. (2022). *Cloud Computing: An Industrial Approach.* Lap Lambert Academic Publishing, ISBN: 978-620-49680-0.
4. Shahnawaz Ahmad, M. A., Farhana Mariyam. (2022). *Fundamentals of Machine Learning for Predictive Data Analytics.* Lap Lambert Academic Publishing, ISBN: 978-60-7-45520-1.

Conference Proceedings

1. Chaturvedi, P., Ahmad, S., & Mewada, A. (2026). Energy-efficient green AI-driven task scheduling for sustainable fog computing in IoT ecosystems. *2026 IEEE 15th International Conference on Communication Systems and Network Technologies (CSNT)*, pp. 1057–1062. IEEE.
2. Chaturvedi, P., Ahmad, S., & Mewada, A. (2026). Fog computing for intelligent and sustainable IoT: Architectures, optimization, datasets, and open challenges. *2026 IEEE Madhya Pradesh Section Conference (MPCON)*, pp. 1178–1183. IEEE.
3. Pandey, D., Chaurasia, A., & Ahmad, S. (2026). Real-Time Student Engagement Detection in Classrooms Using Deep Learning-Based Facial Emotion Recognition. *2026 6th International Conference on Multimedia Signal Processing and Communication Technologies Impact 2026*.
4. Ansari, M. A., Gupta, K., Ahmad, S., Kumar, L., & Mewada, A. (2026). Multiscale Unified Bottleneck Architecture with Spatial Hierarchical Refinement and Aggregation Network for Satellite Image Classification. *Procedia Computer Science*, 283, 993–1002.
5. Bokhari, M. U., Ansari, S. A., Afzal, S., & Ahmad, S. (2026). A Novel Lightweight Stream Cipher Based on NLFSRs and Chaos Function. *2026 2nd International Conference on Cognitive Computing in Engineering Communications Sciences and Biomedical Health Informatics Ic3ecsbhi 2026*, 864–868.
6. Pandey, D., Ahmad, S., Ansari, M. A., & Anas, M. (2026). A SIHO-Optimized Distributed CNN for Secure Parkinson's Disease Prediction in IoT Enabled Healthcare with Social-Media Enhanced Patient Interaction. *Proceedings of the 2026 13th International Conference on Computing for Sustainable Global Development Indiacom 2026*.
7. Pandey, D., Ahmad, S. et al. (2026). Hybrid AI Models for Improved Bone Fracture Diagnosis. *2026 2nd International Conference on Cognitive Computing in Engineering Communications Sciences and Biomedical Health Informatics Ic3ecsbhi 2026*, 541–546.
8. Ahmad, S., Arif, M., Ansari, M. A., Mewada, A., & Raju. (2025). Towards Trustworthy and Scalable IoT: A Blockchain-Federated Learning Approach Based on GSR-C2N. *2025 IEEE Madhya Pradesh Section Conference Mpcon 2025*, 578–582.
9. Ansari, M. A., Ahmad, S., Mewada, A., & Maurya, S. K. (2025). PosturalNet, an Advanced Architecture to Detect Shoplifting at Megastores. *2025 IEEE 14th International Conference on Communication Systems and Network Technologies Csnt 2025*, 863–867.
10. Mewada, A., Sachan, R., Ansari, M. A., & Ahmad, S. (2025). Enhanced Image Steganography: A Secure Approach for Text-in-Image and Image-in-Image Encryption. *2025 IEEE Madhya Pradesh Section Conference Mpcon 2025*, 531–536.
11. Ahmad, S., Husain, M., Ansari, M. A., Arif, M., & Mewada, A. (2025). A Graph Attention-Driven Boosted Deep Learning Framework for Early Parkinson's Detection from Hand-Drawing Biomarkers. *2025 IEEE Madhya Pradesh Section Conference Mpcon 2025*, 84–88.

12 Mewada, A., Maurya, S. K., Ansari, M. A., Sharma, O. P., Avdhesh Yadav, S., & Ahmad, S. (2025). Deceptive Opinion Detection Using Stacking-Based Deep Ensemble Learning. *Icdt 2025 3rd International Conference on Disruptive Technologies*, 1614–1617.

13 Mewada, A., Yadav, A. S., Singh, H. P., Nigam, P., Ansari, M. A., & Ahmad, S. (2025). Smart Censorship: Real-Time Obscene Content Detection Using Skin Pixel Aggregation and Humanoid Shape Recognition. *2025 IEEE Madhya Pradesh Section Conference Mpcon 2025*, 743–747.

14 Ansari, M. A., Upadhyay, S., Mewada, A., Ahmad, S., & Lekhraj. (2025). MPFAN: Multi-Scale Parallel Feature Aggregation Network for Plant Disease Diagnosis. *2025 International Conference on Engineering Innovations and Technologies Icoeit 2025*, 1478–1483.

15 Ansari, S. A., Ali, S., Luqman, M., & Ahmad, S. (2025). Deep Learning Enabled Secure IoT Module for Smart Agriculture in Diverse Environmental Conditions. *2025 IEEE 14th International Conference on Communication Systems and Network Technologies Csnt 2025*, 94–99.

16 Ahmad, J., Ahmad, S., Arif, M., & Nazim, M. (2024). Cloud computing for drug discovery and telemedicine in QSAR analysis. *2024 13th International Conference on System Modeling & Advancement in Research Trends (SMART)*, pp. 225–229. IEEE.

17 Ahmad, S., Ansari, M. A., Mair, A., & Hussain, S. (2024). Smart market insights: A deep learning approach to trend analysis. *2024 13th International Conference on System Modeling & Advancement in Research Trends (SMART)*, pp. 548–555. IEEE.

18 Ahmad, S., Hussain, S., Anwar, K., Verma, H., Sharma, D., & Aggarwal, A. (2024). Exploring the landscape of cloud robotics: A comprehensive review. *2024 15th International Conference on Computing Communication and Networking Technologies (ICCCNT)*, pp. 1–7. IEEE.

19 Ahmad, S., Hussain, S., Wasid, M., Onyarin, O. J., Arif, M., & Ahmad, J. (2024). The future of recruitment: Using deep learning to build intelligent interview bots. *2024 15th International Conference on Computing Communication and Networking Technologies (ICCCNT)*, pp. 1–6. IEEE.

20 Ahmad, S., Lakshmi, T. V., Arunkumar, K., Ravivarman, G., Senthilkumar, G. et al. (2024). IoT-enabled smart e-healthcare system with predictive prescription algorithm for automatic patient monitoring and treatment. *2024 4th International Conference on Pervasive Computing and Social Networking (ICPCSN)*, pp. 1076–1081. IEEE.

21 Ahmad, S., Mehfuz, S., & Beg, J. (2024). Service centric attribute based block level two step encryption scheme for improved cloud security. *AIP Conference Proceedings*, Vol. 2816. AIP Publishing.

22 Anas, M., Hussain, S., & Ahmad, S. (2024). Secure cloud-based diabetes classification with machine learning. *2024 International Conference on Artificial Intelligence and Emerging Technology (Global AI Summit)*, pp. 900–905. IEEE.

23 Arif, M., Ahmad, S., Nazim, M., Ahmad, J., & Onyarin, O. J. (2024). Unveiling the nexus: Exploring the societal implications of AI-driven cybersecurity and data analysis. *2024 13th International Conference on System Modeling & Advancement in Research Trends (SMART)*, pp. 119–128. IEEE.

24 Chauhan, R., Ahmad, S., Onyarin, O. J., & Kumar, R. (2024). Intelligent deep learning model for securing KMS through CSF deviation detection. *2024 13th International Conference on System Modeling & Advancement in Research Trends (SMART)*, pp. 135–143. IEEE.

25 Hussain, S., Ahmad, S., Wasid, M., Chawla, S., Sangwan, A. et al. (2024). Intelligent learning models for cancer diagnosis and treatment: A comprehensive analysis. *2024 7th International Conference on Contemporary Computing and Informatics (IC3I)*, Vol. 7, pp. 1220–1224. IEEE.

26 Nazim, M., Ahmad, S., Arif, M., Mishra, V., & Ahmad, J. (2024). Optimizing software requirements selection: A hybrid k-means and intuitionistic fuzzy TOPSIS approach. *2024 13th International Conference on System Modeling & Advancement in Research Trends (SMART)*, pp. 750–755. IEEE.

27 Padhy, S., Lakshmi, T. V., Ahmad, S., Sophia, S., Senthilkumar, G., & Natrayan, L. (2024). Enhancing IoT-enabled healthcare with genetic-based encryption and authentication for secure and efficient wireless data transmission. *2024 International Conference on Inventive Computation Technologies (ICICT)*, pp. 1873–1878. IEEE.

28 Wasid, M., Ahmad, S., Hussain, S., & Ansari, M. S. (2024). Deep learning based object detection methods: Current status and future prospects. *2024 7th International Conference on Contemporary Computing and Informatics (IC3I)*, Vol. 7, pp. 1213–1219. IEEE.

29 Anwar, K., Garg, A., Ahmad, S., & Wasid, M. (2024). Machine Learning Models for Sentiment Analysis: State of the Art and Applications. *2024 15th International Conference on Computing Communication and Networking Technologies Icccnt 2024*.

30 Ahmad, J., Ahmad, S., Arif, M., & Nazim, M. (2023). Gaining perspective into cloud computing: A business-oriented approach. *2023 International Conference on Computing, Communication, and Intelligent Systems (ICCCIS)*, pp. 1035–1044. IEEE.

31 Kashif, M., Mehfuz, S., Shakeel, I., & Ahmad, S. (2023). Employing an ECC-based hybrid data encryption method to improve multitenancy security in cloud computing. *2023 International Conference on Recent Advances in Electrical, Electronics & Digital Healthcare Technologies (REEDCON)*, pp. 79–83. IEEE.

32 Kashif, M., Shakeel, I., Ahmad, S., & Mehfuz, S. (2023). Enhanced pixel privacy: Leveraging deep learning for advanced image steganography. *2023 7th International Conference on I-SMAC (IoT in Social, Mobile, Analytics and Cloud)*, pp. 535–541. IEEE.

33 Shakeel, I., Mehfuz, S., & Ahmad, S. (2023). Implementing a serverless workflow using AWS Step Function. *2023 International Conference on Recent Advances in Electrical, Electronics & Digital Healthcare Technologies (REEDCON)*, pp. 68–73. IEEE.

34 Shakeel, I., Mehfuz, S., & Ahmad, S. (2022). Securing data in cloud: Major threats and recent strategies. *2022 11th International Conference on System Modeling & Advancement in Research Trends (SMART)*, pp. 1602–1607. IEEE.

35 Ahmad, S., Mehfuz, S., & Beg, J. (2021). Fuzzy cloud access security broker for requirements negotiation and prioritization. *International Conference on Power Electronics, Control and Automation (ICPECA)*, pp. 1–6.

36 Ahmad, S., Mehfuz, S., & Beg, J. (2021). Fuzzy TOPSIS-based cloud model to evaluate cloud computing services. *Advances in Mechanical Engineering (G. Manik, S. Kalia, S. K. Sahoo, T. K. Sharma, & O. P. Verma, Eds.)*, pp. 37–52. Singapore: Springer Singapore.

37 Ahmad, S., Mehfuz, S., & Beg, J. (2021). Enhancing Security of Cloud Platform with Cloud Access Security Broker. *Lecture Notes in Networks and Systems*, 190, 325–335.

38 Ahmad, S., Mehfuz, S., & Beg, J. (2020). Securely work from home with CASB policies under COVID-19 pandemic: A short review. *2020 9th International Conference System Modeling and Advancement in Research Trends (SMART)*, pp. 109–114. IEEE.

Book Chapters

1 Ahmad, S., Hussain, S., Ansari, M. A., & Kumar, N. (2026). *AI-driven healthcare optimization: Integrating fog and edge computing for secure and real-time medical solutions.* Springer.

2 Golash, R., Ahmad, S., Bhawana, & Kumar, N. (2026). *BC and federated learning for the management and orchestration of network slices in 5G, fog, edge, and clouds.* Springer.

3 Mewada, A., Ansari, M. A., Ahmad, S., Maurya, S. K., & Kumar, L. (2026). *IDSCI: An advanced machine learning-based intrusion detection system for cloud infrastructure.* IGI Global Scientific Publishing.

4. Nazim, M., Ahmad, S., Ansari, M. A., & Mewada, A. (2026). *Using large language models to software requirements selection for scalable, explainable, and reliable results.* IGI Global Scientific Publishing.
5. Ansari, M. A., Ahmad, S. et al. (2026). *Raindrop Removal in Images and Videos Using Generative AI: A Survey.* AI Generated Image and Video Synthesis Deep Learning Models Applications and Ethical Implications in Visual Media Creation, 59–72.
6. Kumar, L., Ahmad, S. et al. (2026). *Content-aware Convolutional VAE for Anime Face Synthesis.* AI Generated Image and Video Synthesis Deep Learning Models Applications and Ethical Implications in Visual Media Creation, 123–134.
7. Mewada, A., Ansari, M. A., Jayaswal, R., Ahmad, S., Singh, D. K., Singh, N., Dewang, R. K. et al. (2026). *Nutrition Meets AI: Opportunities in Recommender Systems for Healthier Choices.* Advancing Equity in Global Healthcare Systems, 401–433.
8. Ahmad, S., Wasid, M., Hussain, S., Shahab, S., & Mustafa, J. (2025). *Leveraging the Internet of Things and Blockchain for Developing Intelligent Learning Models.* Applications of Blockchain Technology, 90–103.
9. Ahmad, S., Ansari, M. A., Mewada, A., Raju, R., & Upadhyay, S. (2025). *BioCrypt: A Bioinformatics-Driven Approach to Secure Cloud Computing.* AI Driven Security and Intelligence in Cloud and Internet of Things Systems, 297–328.
10. Ansari, M. A., Mewada, A., Ahmad, S., Kumar, L., Maurya, S. K., Naz, R., & Malviya, L. (2025). *Decoding human activities: Algorithms, frameworks, and challenges in recognition systems.* Neural Network Advancements in the Age of AI, 401–432.
11. Shahnawaz Ahmad, S. M., & Beg, J. (2023). *Empirical impact of AI and IoT on performance management: A review.* Video Data Analytics for Smart City Applications: Methods and Trends, Bentham Science, Vol. 01, pp. 99–111, ISBN: 978-981-5123-70-8.
12. Shahnawaz Ahmad, S. M., & Beg, J. (2022). *Empirical analysis of security enabled quantum computing for cloud environment.* Quantum, Blockchain for Modern Computing Systems: Vision, and Advancements, Vol. 133. Springer, Cham.

Awards

2026 **World's Top 2% Scientist:** Listed among the world's top 2% scientists by Stanford University.

2024 **Research Design Innovation Award:** Awarded by Bennett University during the Dean's Research and Excellence Award Ceremony 2023-2024, held at Bennett University.

2019 **Best Faculty Award:** Awarded by Mewat Engineering College, Nuh, at Capacity Building and Energy Conservation Workshop on Energy Conservation.

Professional Experiences

2025 **Session Chair:** Recent Trends in Renewable Energy and Advancement in Engineering and Technology (RTREAET-2025), Mewat Engineering College (WAQF), Technically Sponsored by Haryana Renewable Energy Development Agency (HAREDA) Section.

2024 **Co-Session Chair:** Global AI Summit 2024, An IEEE Conference Organised by the School of Computer Science Engineering and Technology, Bennett University, Greater Noida, India.

2023 **Track Chair:** Recent Trends in Renewable Energy and Advancement in Engineering and Technology (RTREAET-2023), Mewat Engineering College (WAQF), Technically Sponsored by Haryana Renewable Energy Development Agency (HAREDA) Section.

Skills

Programming	C · C++ · Python · PHP
Web Development	HTML · CSS
Server	Xampp
Cloud Applications	Amazon AWS EC2
Simulators	CloudSim
Database	MySQL · MongoDB
Software Packages	Matlab · LaTeX

Online Courses

New York University	Cyber Attack Countermeasures · Real-Time Cyber Threat Detection and Mitigation · Enterprise and Infrastructure Security · Introduction to Cyber Attacks
University of California, Irvine	Introduction to Project Management Principles and Practices · The Arduino Platform and C Programming · Introduction to Internet of Things and Embedded Systems · Project Management · Managing Project Risk and Changes · Initiating and Planning Projects · Budgeting and Scheduling Projects
Duke University	Introduction to Machine Learning · Programming Fundamentals
University of Michigan	Introduction to HTML 5 · Programming for Everybody (Getting Started with Python)
Amazon Web Services	AWS Fundamentals: Going Cloud Native · Getting Started with AWS Machine Learning
Stanford University	Cryptography I
University of Illinois	Cloud Computing Applications, Part 1: Cloud Systems and Architecture
Google Cloud	Introduction to Cloud Identity
University of California San Diego	Introduction to Big Data
Arizona State University	Data Management in the Cloud
University of California	Introduction to Search Engine Optimization
University of London	Get Interactive: Practical Teaching with Technology
University of Toronto	Introduction to GIS Mapping
The State University of New York (SUNY)	How to Write a Resume (Project-Centered Course)
LearnQuest, Coursera	Developing AI Applications on Azure
Udemy	Fundamentals of Cloud Computing

Summer School/Short-term/Refresher Courses

8 Weeks	**Summer Faculty Research Fellow Programme** from 16 May 2023 - 14 July 2023, hosted by *Indian Institute of Technology (IIT), Delhi*.
4 Weeks	**Induction/Orientation Programme** from June 04 - July 01, 2020, organized by the Teaching Learning Centre, Ramanujan College, University of Delhi under the aegis of Ministry of Education, Govt. of India.

2 Weeks **Refresher Course in Computational and Mathematical Sciences** from October 22nd - November 05th 2021, organized by the UGC-HRDC, Jamia Millia Islamia, New Delhi.

1 Week **Faculty Development Programme on Data Analytics Using Python** from 3rd June - June 07th 2024, conducted by ICT Academy (Govt. of India) at Bennett University, Greater Noida, India.

1 Week **Short Term Course on Outcome Based Curriculum** from June 25 - June 29, 2018, conducted by Curriculum Development Centre at NITTTR, Chandigarh.

Reviewer

- The Journal of Supercomputing, Springer
- Scientific Reports, Springer Nature
- Signal, Image and Video Processing, Springer
- Discover Computing, Springer
- Peer-to-Peer Networking and Applications, Springer
- Journal of Big Data, Springer
- Evolutionary Intelligence, Springer
- Knowledge and Information Systems, Springer
- Earth Science Informatics, Springer
- IEEE Access, IEEE
- Future Generation Computer Systems, Elsevier
- Alexandria Engineering Journal
- Sustainable Computing: Informatics and Systems, Elsevier
- Concurrency and Computation: Practice and Experience, Wiley
- Multimedia Tools and Applications, Springer
- Cyber Security and Applications, Elsevier
- SN Computer Science, Springer
- International Journal of System Assurance Engineering and Management, Springer
- International Journal of Electrical Engineering
- Bulletin of Electrical Engineering and Informatics

References

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